

Personal Notes on Tensor Calculus

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The Outline

These notes will develop and motivate the intrinsic calculus of tensor fields over smooth manifolds alongside related concepts from differential geometry. Prerequisites include, but are not limited to set theory, linear algebra, and vector calculus. The Einstein summation convention will be utilized throughout.

Topological Spaces

In order to establish the notion of a manifold, some groundwork needs to be established first. Let $M \neq \emptyset$ be a collection of elements. If $p \in M$, then p is said to be a point in M . Furthermore, $\tau \subseteq \mathcal{P}(M)$ is said to be a topology of M if it satisfies the following properties (axioms):

- $\emptyset, M \in \tau$ ————— Nothing & Everything —————
- $\forall I \subseteq \mathbb{N}, \{U_i\}_{i \in I} \subseteq \tau \implies \bigcup_{i \in I} U_i \in \tau$ ————— Closure Under Unions —————
- $\forall n \in \mathbb{N}, \{U_i\}_{i=1}^n \subseteq \tau \implies \bigcap_{i=1}^n U_i \in \tau$ ————— Closure Under Finite Intersections —————

The pairing (M, τ) is said to be a topological space, and any element of τ is said to be an open subset of M by definition. Now, given any $p \in M$, let $\mathcal{N}(p) = \{U \in \tau \mid p \in U\}$ denote the neighborhood system of the point p . If $U \in \mathcal{N}(p)$, then U is called a neighborhood of p . The following notions can now be formalized:

- p is an interior point of $S \subseteq M$ (i.e. $p \in \text{Int}(S)$) if $\exists U \in \mathcal{N}(p)$ such that $U \subseteq S$.
- p is an exterior point of $S \subseteq M$ (i.e. $p \in \text{Ext}(S)$) if $\exists U \subseteq \mathcal{N}(p)$ such that $U \subseteq M \setminus S$.
- p is a boundary point of $S \subseteq M$ (i.e. $p \in \partial S$) if $\forall U \in \mathcal{N}(p), U \cap S \neq \emptyset$ and $U \cap (M \setminus S) \neq \emptyset$.
- p is an accumulation point of $S \subseteq M$ (i.e. $p \in S'$) if $\forall U \in \mathcal{N}(p), U \setminus \{p\} \cap S \neq \emptyset$.

The closure of any such $S \subseteq M$ is thus defined by $\text{Cl}(S) = \text{Int}(S) \cup \partial S$, and without proof, it can be shown that $\text{Cl}(S) = S \cup S'$. It is said that S is closed if $S = \text{Cl}(S)$. It likewise follows that S is open if $S = \text{Int}(S)$. These definitions clearly partition all the regions of M with respect to any one of its subsets.

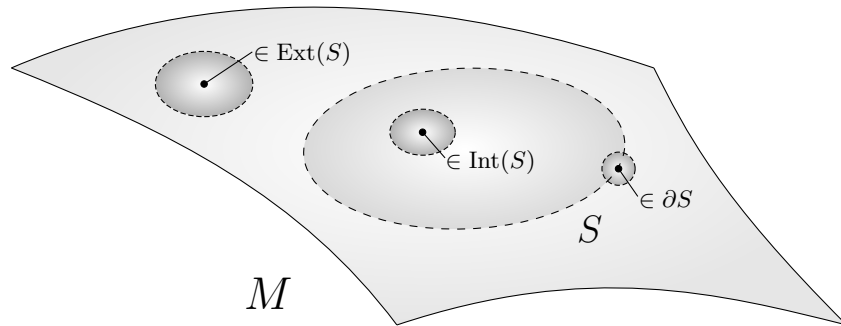


Figure 1: Region Partitioning

By definition, a point $p \in M$ is said to be close (adherent) to a set $S \subseteq M$ if $p \in \text{Cl}(S)$. In the same respect, the points $p, q \in M$ are said to be close to each other if $q \in U$ for every $U \in \mathcal{N}(p)$, or equivalently, $p \in V$ for every $V \in \mathcal{N}(q)$. In order to make this notion more precise, a metric would have to be imposed.

The Hausdorff Property

With the established structure at hand, (M, τ) is considered Hausdorff if $\forall p \neq q, \exists U \in \mathcal{N}(p)$ and $\exists V \in \mathcal{N}(q)$ such that $U \cap V = \emptyset$. In other words, any two distinct points in M can be located respectively in their own neighborhoods, which do not overlap with each other.

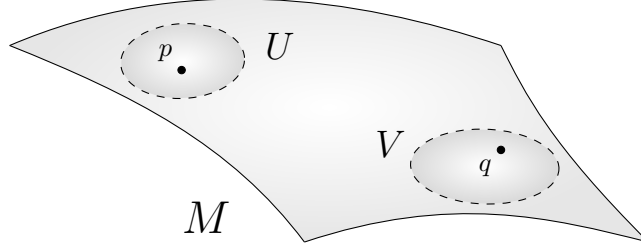


Figure 2: Distinct Neighborhoods

Let $s : \mathbb{N} \rightarrow M$ be a sequence. By definition, s converges to $p \in M$ (i.e. $s \rightarrow p$) if $\forall U \in \mathcal{N}(p), \exists m \in \mathbb{N}$ such that $n \geq m \implies s_n \in U$. Intuitively, no matter how small the neighborhood U is around p , there always exists a place in the sequence s beyond which the outputs of s reside entirely inside U .

Assuming (M, τ) is Hausdorff, suppose that the sequence s converges to both $p, q \in M$. There necessarily exists open sets $U, V \subset M$ such that $p \in U$ and $q \in V$. It directly follows that $\exists m \in \mathbb{N}$ such that $n \geq m \implies s_n \in U \cap V$. Thus, by assumption, if $p \neq q$, then one can choose U, V such that $U \cap V = \emptyset$, which would be a contradiction, implying $p = q$. That is to say convergent limits of sequences are unique.

Topological Basis

Given the topological space (M, τ) , a collection of open subsets $\mathcal{B} \in \tau$ is called a basis of the topology τ if $\forall U \in \tau, \exists \{U_i\}_{i \in I} \subseteq \mathcal{B}$ such that $\bigcup_{i \in I} U_i = U$, where $I \subseteq \mathbb{N}$.

The topological space (M, τ) is said to be first-countable if $\forall p \in M, \exists U_k \subseteq U \in \mathcal{N}(p)$, with $k \in \mathbb{N}$, such that $\mathcal{N}(p) = \{U_k\}_{k \in I}$, where $I \subseteq \mathbb{N}$. In other words, in this context, every neighborhood system can be generated by a countable set of neighborhoods. This set is referred to as a countable neighborhood basis.

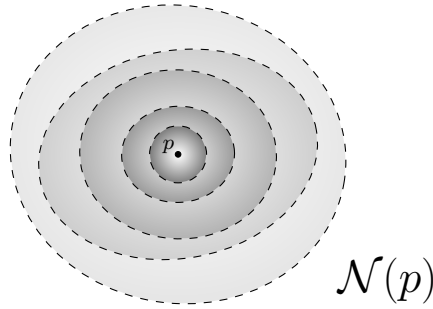


Figure 3: Countable Neighborhood Basis

Provided that $I \subseteq \mathbb{N}$, the topological space (M, τ) is said to be second-countable if $\exists \mathcal{B}_k \in \tau$, where $k \in \mathbb{N}$, such that $\mathcal{B} = \{\mathcal{B}_k\}_{k \in I}$ is a basis of τ . In short, $\mathcal{B}$ can be generated by a countable set of open sets. It can be shown that (M, τ) being second-countable implies that the space is first-countable.

Suppose (M, τ) is second-countable, which implies (contextually) that $\mathcal{B} = \{\mathcal{B}_k\}_{k \in I}$ is a basis of τ . For any $p \in M$, let $U \in \mathcal{N}(p)$ and $\mathcal{B}_p = \{V \in \mathcal{B} \mid p \in V\}$, which is countable. At the same time, since $\mathcal{B}$ is a basis, $\exists \mathcal{B}_k \in \mathcal{B}_p$ such that $\mathcal{B}_k \subseteq U$. Therefore, $\mathcal{B}_p$ generates U , implying the space is first-countable.

Topological Continuity

Suppose (M, τ_M) and (N, τ_N) are topological spaces. The map $F : M \rightarrow N$ is called continuous (with respect to τ_M and τ_N) if $V \in \tau_N \implies F^{-1}(V) \in \tau_M$, where F^{-1} is the preimage of F . That is to say, for any $p \in M$, fixing the neighborhood $V \in \mathcal{N}(F(p))$ guarantees the existence of another neighborhood $U \in \mathcal{N}(p)$ such that $F(U) \subseteq V$.

Intuitively, assuming that F is continuous, regardless of how small of a neighborhood V is chosen around the output $F(p)$, a sufficiently small neighborhood U exists around p that is mapped to the inside of V .

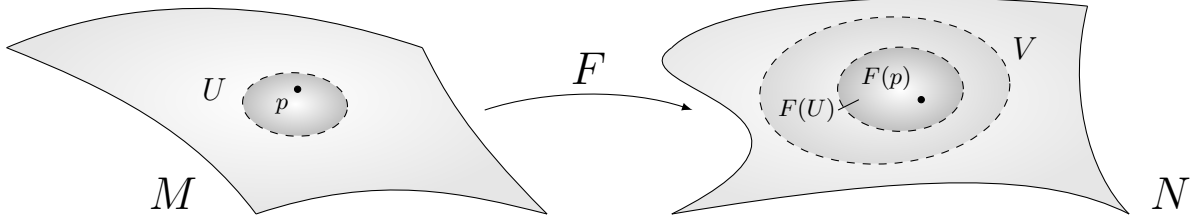


Figure 4: Continuous Mapping

If $F : M \rightarrow N$ is continuous and bijective, and F^{-1} is continuous, then F is called a homeomorphism. Moreover, if $\forall p \in M, \exists U \in \mathcal{N}(p)$ such that $F(U) \in \tau_N$, then $F|_U$ is called a local homeomorphism. It is said that a pair of topological spaces are (locally) homeomorphic if there exists a (local) homeomorphism between them.

Now, provided that $s : \mathbb{N} \rightarrow M$ is any sequence, another sequence derived via composition can be built as given by $F(s) : n \in \mathbb{N} \mapsto F(s_n)$. If $s \rightarrow p \implies F(s) \rightarrow F(p)$, then F is said to be sequentially continuous.

Suppose F is continuous and $s \rightarrow p$, and let $V \in \mathcal{N}(F(p))$. It follows from continuity that $p \in F^{-1}(V) \in \tau_M$. By supposition, $\exists m \in \mathbb{N}$ such that $n \geq m \implies s_n \in F^{-1}(V)$, which further implies $F(s_n) \in V$, and thus $F(s) \rightarrow F(p)$. More concisely, continuity implies sequential continuity.

Provided that (M, τ_M) is first countable and $F : M \rightarrow N$ is sequentially continuous, fix $p \in M$. Let $\{U_k\}_{k \in \mathbb{N}}$ be a countable neighborhood basis at p such that $U_{k+1} \subseteq U_k$ for all k , and let $V \in \mathcal{N}(F(p))$. For each $k \in \mathbb{N}$, either $F(U_k) \subseteq V$, in which case $\exists U_k \in \mathcal{N}(p)$ satisfying $F(U_k) \subseteq V$, or $\exists s_k \in U_k$ such that $F(s_k) \notin V$. In the latter case, for all $k \in \mathbb{N}$, define a sequence $s : n \in \mathbb{N} \mapsto s_n \in U_n$. Given any $U \in \mathcal{N}(p)$, $\exists m \in \mathbb{N}$ such that $U_m \subseteq U$, hence $s_n \in U$ for all $n \geq m$, implying $s \rightarrow p$. By sequential continuity, $F(s) \rightarrow F(p)$, and since $V \in \mathcal{N}(F(p))$, $\exists m' \in \mathbb{N}$ such that $n \geq m' \implies F(s_n) \in V$. Therefore, for $k = m'$, there does not exist any $q \in U_k$ such that $F(q) \notin V$, and thus $F(U_k) \subseteq V$. Since V was arbitrary, F is continuous at p . In other words, sequential continuity of F implies continuity of F in a first-countable space.

Metric Spaces

From scratch, let $M \neq \emptyset$ be a collection of elements. Suppose M admits a distance function $d : M \times M \rightarrow \mathbb{R}$. If the function d has the following properties, then it is a metric, and (M, d) is called a metric space:

- $\forall x, y \in M, d(x, y) \geq 0$ ————— Strict Non-negativity —————
- If $x, y \in M$, then $d(x, y) = 0$ iff $x = y$ ————— Indistinguishability —————
- $\forall x, y \in M, d(x, y) = d(y, x)$ ————— Symmetrical —————
- $\forall x, y, z \in M, d(x, y) \leq d(x, z) + d(y, z)$ ————— Triangle Inequality —————

Intuitively, the metric quantifies how close any two points are with respect to a specific distance convention. Given this structure, one can construct a definition for open balls at points. In particular, provided that $\varepsilon > 0$, the open ε -ball at $x \in M$ is the set given by $B_\varepsilon(x) = \{y \in M \mid d(x, y) < \varepsilon\}$.

Properties of Metric Spaces

With a metric space at hand, a topology on M can be constructed. Specifically, upon defining the set $\tau_d = \{U \subseteq M \mid \forall x \in U, \exists \epsilon > 0 : B_\epsilon(x) \subseteq U\}$, it can be identified as a topology. That is to say, given $U \subseteq M$, if all points in U are individually equipped with at least one ball that resides entirely inside U , then the subset U can be considered to be open. The proof is straight forward:

- By inspection, $\emptyset \in \tau_d$. Likewise, for any $x \in M$, choose $\epsilon > 0$. It follows that $B_\epsilon(x) \subset M$, which implies $M \in \tau_d$. Therefore, it consequently holds that $\emptyset, M \in \tau_d$.
- Contextually, let $\{U_i\}_{i \in I} \subseteq \tau_d$, and define $U = \bigcup_{i \in I} U_i$. Take $x \in U$. There necessarily exists some $k \in I$ such that $x \in U_k \in \tau_d$. Thus, $\exists \epsilon > 0$ with $B_\epsilon(x) \subseteq U_k \subseteq U$, which implies $U \in \tau_d$.
- Let $U_1, U_2 \in \tau_d$, and take $x \in U_1 \cap U_2$, which implies $x \in U_1, U_2$. There necessarily exists $\epsilon_1, \epsilon_2 > 0$ such that $U_1 \subseteq B_{\epsilon_1}(x)$ and $U_2 \subseteq B_{\epsilon_2}(x)$. Take $\epsilon = \min(\epsilon_1, \epsilon_2)$, which implies $B_\epsilon(x) \subseteq U_1 \cap U_2$, further implying $U_1 \cap U_2 \in \tau_d$.

Consequently, every metric space induces a topological space. Now, consider two points $x \neq y$, and thus $d(x, y) > 0$. Let $\epsilon = \frac{1}{3}d(x, y)$, and suppose $z \in B_\epsilon(x) \cap B_\epsilon(y)$. It follows that $d(x, z), d(y, z) < \epsilon$, and so by the triangle inequality, $d(x, y) \leq d(x, z) + d(y, z) < 2\epsilon = \frac{2}{3}d(x, y)$, which is impossible. Therefore, $B_\epsilon(x) \cap B_\epsilon(y) = \emptyset$, which implies (M, τ_d) is Hausdorff.

Next, define the (countable) unit ball system at $x \in M$ to be $\mathcal{N}_B(x) = \{B_{1/n}(x) \mid n \in \mathbb{N}\}$, in which it follows that $\mathcal{N}_B(x) \subseteq \tau_d$. Suppose $U \in \mathcal{N}(x)$, and note $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq U$. Likewise, $\exists n \in \mathbb{N}$ such that $1/n < \epsilon$, which implies $B_{1/n}(x) \subseteq B_\epsilon(x)$. Therefore, $B_{1/n}(x) \subseteq U$, and thus $\mathcal{N}_B(x)$, which is countable, generates $\mathcal{N}(x)$. It can now be concluded (M, τ_d) is first-countable.

Euclidean Space

For some $n \in \mathbb{N}$, consider the collection of n -tuples $\mathbb{R}^n = \{x = (x_1, \dots, x_n) \mid x_i \in \mathbb{R}, i \in \mathbb{N}_n\}$. Using a metric, a notion of distance between points can be established. In particular, if $x = (x_1, \dots, x_n)$, by definition, $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$, which is the standard (Euclidean) norm on $\mathbb{R}^n$. Upon inspection, the distance function given by $d : (x, y) \in \mathbb{R}^n \times \mathbb{R}^n \mapsto \|x - y\|$ is a valid metric on $\mathbb{R}^n$. Therefore, $(\mathbb{R}^n, d)$ is a metric space which induces the Hausdorff first-countable topological space $(\mathbb{R}^n, \tau_d)$.

Now, construct the basis of $\mathbb{R}^n$ to be $\mathcal{B}_d = \{B_\epsilon(q) \mid q \in \mathbb{Q}^n, \epsilon \in \mathbb{Q}^+\}$. Suppose $U \subset \mathbb{R}^n$ is open, and let $x \in U$. Then $\exists \epsilon > 0$ such that $B_\epsilon(x) \subseteq U$. Choose $q \in \mathbb{Q}^n$ such that $d(x, q) < \epsilon/2$, and take $r \in \mathbb{Q}^+$, where $r < \epsilon/2$. Then $B_r(q) \subseteq B_\epsilon(x) \subseteq U$. Because $\mathbb{Q}^n$ is countable, $\mathcal{B}_d$ is countable. Therefore $\mathcal{B}_d$ is a countable basis on τ_d , which implies $(\mathbb{R}^n, \tau_d)$ is second-countable.

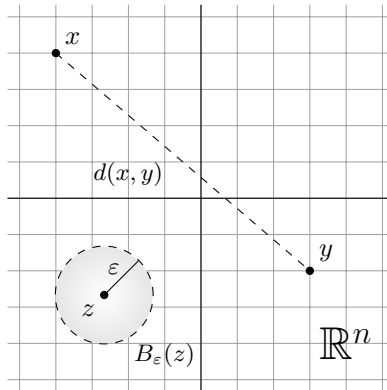


Figure 5: The Euclidean Metric

This space has a basis $\beta_n = \{\hat{x}_i\}_{i=1}^n$ such that $\forall x = (x^1, \dots, x^n) \in \mathbb{R}^n$, one can write (via the Einstein summation convention) $x = x^i \hat{x}_i$. Since β_n is linearly independent, there exists a map $(x^1, \dots, x^n) \mapsto x^i$.

Local Coordinate Charts

Contextually, if there exists a local homeomorphism $\varphi : U \subseteq M \rightarrow \mathbb{R}^n$, which establishes a coordinate system on M , then U and $\mathbb{R}^n$ are locally homeomorphic, and M is locally Euclidean on U by definition. To be precise, the pairing (U, φ) is referred to as a (local) coordinate chart on M .

Assume (apriori) that M is locally homeomorphic to $\mathbb{R}^n$ and $\mathbb{R}^m$ ($n, m \in \mathbb{N}$), with respective coordinate charts φ and ψ . Upon fixing $p \in M$, $\exists U, V \in \mathcal{N}(p)$. Provided that $W = U \cap V$, it follows that $\varphi|_W$ and $\psi|_W$ are local homeomorphisms onto open sets in N . Thus, the map $\psi \circ \varphi^{-1} : \varphi(W) \rightarrow \psi(W)$ is a local homeomorphism (bijection). Therefore, $\varphi(W) \cong \psi(W)$, which implies $\mathbb{R}^n \cong \mathbb{R}^m$, further implying $n = m$.

This uniqueness motivates the following definition. If M is locally homeomorphic to $\mathbb{R}^n$, then it is locally Euclidean of dimension n (i.e. $\dim(M) = n$). Note that in the literature, $\dim(M)$ has a compatible, but more primitive definition in which $\dim(M) = n \not\Rightarrow M$ is homeomorphic to $\mathbb{R}^n$.

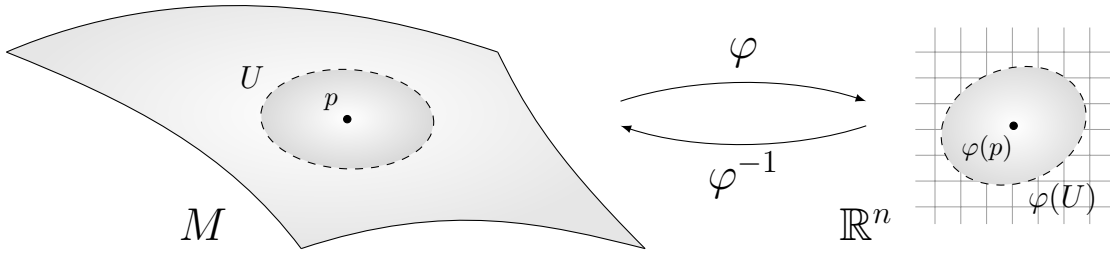


Figure 6: Local Coordinate Chart Mapping

If M is locally Euclidean of dimension n , for any respective (local) coordinate chart $\varphi : M \rightarrow \mathbb{R}^n$, there exists continuous (local) coordinate functions $x^i : M \rightarrow \mathbb{R}$ such that $\varphi : p \in M \mapsto (x^1(p), \dots, x^n(p))$. In this respect, for any $p \in M$, the i^{th} coordinate of p with respect to the chart φ is the i^{th} component of $\varphi(p)$.

Transition Maps

Let M be locally Euclidean of dimension n equipped with (local) coordinate charts given by $\varphi : U \rightarrow D$ and $\psi : V \rightarrow R$ such that $U \cap V \neq \emptyset$. These charts describe the same region in their own coordinate systems, and because these charts are invertible, their respective coordinate systems can be transformed into each other via transition maps. In particular, provided that $D_\cap = \varphi(U \cap V)$ and $R_\cap = \psi(U \cap V)$, these transition maps can be (explicitly) defined by $\psi \circ \varphi^{-1} : D_\cap \rightarrow R_\cap$ and $\varphi \circ \psi^{-1} : R_\cap \rightarrow D_\cap$.

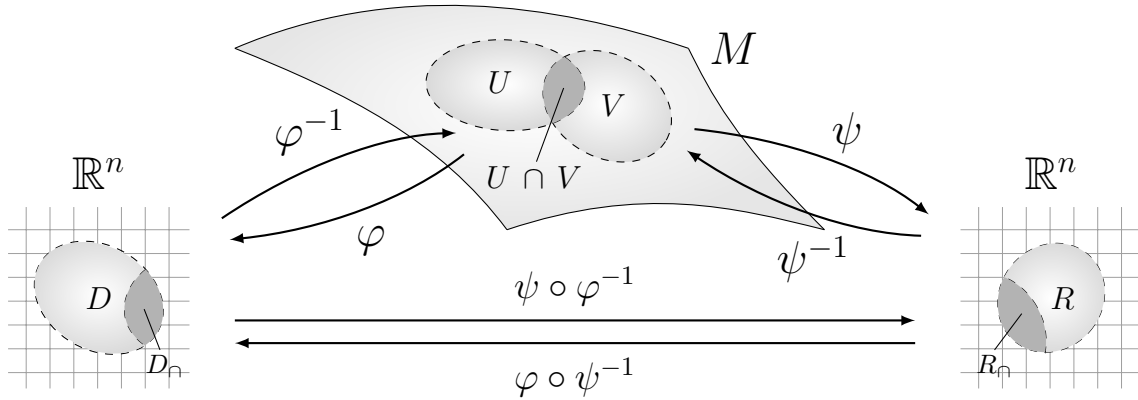


Figure 7: Coordinate Transformation